

Wald Chapter 5: Homogeneous, Isotropic Cosmology

R. M. Wald, General Relativity, Chapter 5; selected worked problems and related solutions.

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General Relativity

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Problem prompts are paraphrased from Robert M. Wald, General Relativity (University of Chicago Press, 1984); the equations, notation, numbering, and guidance follow the source. See the official publisher page.

1. Chapter 5, Problem 1: Robertson–Walker Coordinates

Show that the Robertson–Walker metric (5.1.11) can be rewritten as

$$ds^2 = -d\tau^2 + a^2(\tau) \left[\frac{dr^2}{1 - kr^2} + r^2 (d\theta^2 + \sin^2 \theta d\phi^2) \right]. \quad (1.1)$$

For $k = +1$, determine which part of the three-sphere is covered by these coordinates.

Solution. For $K = 0$, the metric is simply

$$ds^2 = -d\tau^2 + a^2(\tau) d\mathbf{x}^2 = -d\tau^2 + a^2(\tau) (dr^2 + d\Omega_2^2). \quad (1.2)$$

For $K > 0$, we define

$$\sin^2 \psi = r^2, \quad (1.3)$$

which gives

$$\frac{1}{1 - r^2} dr^2 = d\psi^2. \quad (1.4)$$

Therefore, we have

$$ds^2 = -d\tau^2 + a^2(\tau) \left(\frac{dr^2}{1 - r^2} + r^2 d\Omega_2^2 \right). \quad (1.5)$$

Similarly, we find

$$ds^2 = -d\tau^2 + a^2(\tau) \left(\frac{dr^2}{1 + r^2} + r^2 d\Omega_2^2 \right) \quad (1.6)$$

for $k < 0$ by defining $r^2 = \sinh^2 \psi$.

Note that for $K > 0$, $r \in [0, 1]$, which only covers either $\psi \in [0, \pi/2]$ or $\psi \in [\pi/2, \pi]$. Therefore, it only covers the northern or the southern hemisphere of the three-sphere.

2. Chapter 5, Problem 2: The Friedmann Equations for Curved Spatial Sections

Derive Einstein's equations (5.2.14) and (5.2.15) for the three-sphere ($k = +1$) and hyperboloid ($k = -1$) geometries.

Solution. The metric tensor is

$$g_{\mu\nu} = \text{diag}(-1, a^2/(1 - kr^2), a^2r^2, a^2r^2 \sin^2 \theta). \quad (2.1)$$

The Levi-Civita connection is given by

$$\Gamma^\rho_{\mu\nu} = \frac{1}{2}g^{\rho\sigma}(\partial_\mu g_{\nu\sigma} + \partial_\nu g_{\mu\sigma} - \partial_\sigma g_{\mu\nu}). \quad (2.2)$$

We start with

$$\Gamma^i_{i\tau} = \Gamma^i_{\tau i} = \frac{1}{2}g^{ii}(\partial_i g_{ii}) = \frac{\dot{a}}{a}. \quad (2.3)$$

Meanwhile,

$$\Gamma^\tau_{rr} = -\frac{1}{2}g^{\tau\tau}\partial_\tau g_{rr} = \frac{a\dot{a}}{1-kr^2}, \quad \Gamma^\tau_{\theta\theta} = r^2 a\dot{a}, \quad \Gamma^\tau_{\phi\phi} = r^2 \sin^2 \theta a\dot{a}. \quad (2.4)$$

On top of that, we also have

$$\Gamma^r_{rr} = \frac{1}{2}g^{rr}\partial_r g_{rr} = \frac{kr}{1-kr^2}, \quad \Gamma^r_{\theta\theta} = -\frac{1}{2}g^{rr}\partial_\theta g_{rr} = -(1-kr^2)a^2 r, \quad \Gamma^r_{\phi\phi} = -(1-kr^2)a^2 r \sin^2 \theta. \quad (2.5)$$

Finally, we also have

$$\Gamma^\theta_{\phi\phi} = -\frac{1}{2}g^{\theta\theta}\partial_\theta g_{\phi\phi} = -\sin \theta \cos \theta, \quad \Gamma^\theta_{\theta r} = \Gamma^\theta_{r\theta} = \frac{1}{2}g^{\theta\theta}\partial_r g_{\theta\theta} = \frac{1}{r}, \quad \Gamma^\phi_{\phi r} = \Gamma^\phi_{r\phi} = \frac{1}{r}, \quad (2.6)$$

and

$$\Gamma^\phi_{\theta\phi} = \Gamma^\phi_{\phi\theta} = \cot \theta \quad (2.7)$$

with all the rest of the components vanishes.

Now, we are ready to compute the Ricci tensor, which is given by

$$R_{\mu\rho} = \partial_\nu \Gamma^\nu_{\mu\rho} - \partial_\mu \Gamma^\nu_{\nu\rho} + \Gamma^\alpha_{\mu\rho} \Gamma^\nu_{\alpha\nu} - \Gamma^\alpha_{\nu\rho} \Gamma^\nu_{\alpha\mu}. \quad (2.8)$$

$$R_{\tau\tau} = -3\partial_\tau \Gamma^\tau_{rr} - 3\Gamma^r_{rr} \Gamma^\tau_{rr} = -3\frac{\ddot{a}}{a}. \quad (2.9)$$

Note that $R_{**} = \frac{1-kr^2}{a^2} R_{rr}$, and

$$\begin{aligned} R_{rr} &= \partial_\tau \Gamma^\tau_{rr} + \partial_r \Gamma^r_{rr} - \partial_r \Gamma^r_{rr} - 2\partial_r \Gamma^\theta_{\theta r} \\ &\quad + (\Gamma^r_{rr})^2 + 2\Gamma^r_{rr} \Gamma^\theta_{r\theta} + 3\Gamma^r_{rr} \Gamma^\tau_{rr} - (\Gamma^r_{rr})^2 - 2\Gamma^r_{rr} \Gamma^\tau_{rr} - 2(\Gamma^\theta_{\theta r})^2 \\ &= \frac{\dot{a}^2 + a\ddot{a}}{1-kr^2} + \frac{2}{r^2} + \frac{\dot{a}^2}{1-kr^2} + 2\frac{k}{1-kr^2} + \frac{2}{r^2} \\ &= \frac{2\dot{a}^2 + a\ddot{a}}{1-kr^2} + \frac{2k}{1-kr^2}, \end{aligned} \quad (2.10)$$

which gives

$$R_{**} = \left(\frac{\ddot{a}}{a} + 2\frac{\dot{a}^2 + k}{a^2} \right). \quad (2.11)$$

The Ricci scalar is given by

$$R = -R_{\tau\tau} + 3R_{**} = 6 \left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2 + k}{a^2} \right). \quad (2.12)$$

Therefore, we find Einstein's equations to be

$$G_{\tau\tau} = 3\frac{\dot{a}^2 + k}{a^2} = 8\pi\rho, \quad (2.13)$$

$$G_{**} = -2\frac{\ddot{a}}{a} - \frac{\dot{a}^2 + k}{a^2} = 8\pi P. \quad (2.14)$$

Manipulating a bit, we find

$$\frac{3\dot{a}^2}{a^2} = 8\pi\rho - \frac{3k}{a^2}, \quad (2.15)$$

$$\frac{3\ddot{a}}{a} = -4\pi(\rho + 3P). \quad (2.16)$$

3. Chapter 5, Problem 3: Cosmological Constant Solutions

- a) Use the modified Einstein equation (5.2.17) with cosmological constant Λ to write the counterparts of equations (5.2.14) and (5.2.15), including their Λ terms. Show that a static solution exists precisely when $k = +1$ and $\Lambda > 0$. For a dust-filled Einstein static universe ($P = 0$), relate its radius a to the density ρ . Perturb a slightly away from equilibrium and show that the static solution is unstable.

Solution. Einstein's equations become

$$G_{\tau\tau} - \Lambda = 8\pi\rho, \quad (3.1)$$

$$G_{**} + \Lambda = 8\pi P, \quad (3.2)$$

which gives

$$\frac{3\dot{a}^2}{a^2} - \Lambda = 8\pi\rho - \frac{3k}{a^2}, \quad (3.3)$$

$$\frac{3\ddot{a}}{a} - \Lambda = -4\pi(\rho + 3P). \quad (3.4)$$

A static universe would have $\dot{a} = \ddot{a} = 0$, which gives

$$\frac{3k}{a^2} = 8\pi\rho + \Lambda, \quad (3.5)$$

$$\Lambda = 4\pi(\rho + 3P). \quad (3.6)$$

It is easy to see that we must have $k = +1$ for the static universe solution to exist. In the case of a dust-filled static universe, we have $a_0 = \sqrt{1/4\pi\rho}$ and $\Lambda = 4\pi\rho$.

Consider perturbing the universe to $a = a_0 + \delta a$ with $\delta a/a_0 \ll 1$, and release it ($\dot{\delta a} = 0$). Through mass conservation, we have $\rho_0 a_0^3 = \rho(a_0 + \delta a)^3$. We find

$$\ddot{\delta a} = \frac{\delta a}{a_0^2}, \quad (3.7)$$

which means that the static universe is not stable.

- b) Set $\Lambda > 0$ and $T_{ab} = 0$ in the modified Einstein equation. Find the spatially homogeneous and isotropic solution for $k = 0$. This gives de Sitter spacetime; the $k = \pm 1$ solutions correspond to other choices of spacelike hypersurfaces in the same spacetime.

Solution. The equations become

$$\frac{3\dot{a}^2}{a^2} = \Lambda, \quad \frac{3\ddot{a}}{a} = \Lambda. \quad (3.8)$$

This has a solution

$$a(\tau) = Ae^{\sqrt{\Lambda/3}\tau}. \quad (3.9)$$

4. Chapter 5, Problem 4: Cosmological Redshift

Derive the redshift relation (5.3.6) through the following steps.

- a) Show that

$$\nabla_a u_b = \frac{\dot{a}}{a} h_{ab}, \quad \dot{a} = \frac{da}{d\tau}, \quad (4.1)$$

where h_{ab} is defined in equation (5.1.10).

Solution. In a local chart, we have $u^\mu = (1, 0, 0, 0)$. The FRW metric can be written as

$$ds^2 = -d\tau^2 + h_{ij}(\tau)dx^i dx^j. \quad (4.2)$$

The covariant derivative is therefore

$$\nabla_\mu u_\nu = \partial_\mu u_\nu - \Gamma^\rho_{\mu\nu} u_\rho = \Gamma^\tau_{\mu\nu}. \quad (4.3)$$

In problem 5.2, we found

$$\Gamma^\tau_{ij} = \frac{\dot{a}}{a} h_{ij}(\tau), \quad (4.4)$$

where

$$h_{\mu\nu} = \begin{pmatrix} 0 & 0 \\ 0 & h_{ij} \end{pmatrix}. \quad (4.5)$$

Therefore, we find

$$\nabla_\mu u_\nu = \frac{\dot{a}}{a} h_{\mu\nu}(\tau). \quad (4.6)$$

b) Along any null geodesic, prove

$$\frac{d\omega}{d\lambda} = -k^a k^b \nabla_a u_b = -\frac{\dot{a}}{a} \omega^2, \quad (4.7)$$

where λ is an affine parameter.

Solution. Let the photon have coordinate $x^\mu(\lambda)$, then $k^\mu = dx^\mu/d\lambda$. In our case,

$$\frac{d\omega}{d\lambda} = \frac{dx^\mu}{d\lambda} \partial_\mu \omega = k^\mu \nabla_\mu \omega = -k^\mu \nabla_\mu (k^\nu u_\nu). \quad (4.8)$$

Since k^μ satisfies the geodesic equation, we have $k^\mu \nabla_\mu k^\nu = 0$. Therefore, we have

$$\frac{d\omega}{d\lambda} = -\frac{\dot{a}}{a} k^\mu k^\nu h_{\mu\nu} = -\frac{\dot{a}}{a} k^\mu k^\nu (u_\mu u_\nu + g_{\mu\nu}) = -\frac{\dot{a}}{a} \omega^2. \quad (4.9)$$

c) Integrate the result of part (b) to obtain equation (5.3.6).

Solution. Rearranging, we find

$$\frac{d\omega}{\omega^2} = -\frac{d\lambda}{d\tau} \frac{da}{a}. \quad (4.10)$$

Note that

$$\omega = -g_{\mu\nu} \frac{dx^\mu}{d\lambda} \frac{dx^\nu}{d\tau} = -g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} \frac{d\tau}{d\lambda} = \frac{d\tau}{d\lambda}. \quad (4.11)$$

Therefore, we find

$$\frac{d\omega}{\omega} = -\frac{da}{a} \implies \frac{\omega_1}{\omega_2} = \frac{a(\tau_2)}{a(\tau_1)}. \quad (4.12)$$

5. Chapter 5, Problem 5: Radial Null Geodesics and Global Light Travel

Consider a radial null geodesic in a Robertson–Walker cosmology (5.1.11), so that

$$\frac{d\theta}{d\lambda} = \frac{d\phi}{d\lambda} = 0. \quad (5.1)$$

a) For each of the three spatial geometries, show that the change in the coordinate ψ between times τ_1 and τ_2 is

$$\Delta\psi = \int_{\tau_1}^{\tau_2} \frac{d\tau}{a(\tau)}. \quad (5.2)$$

In the flat case ($k = 0$), ψ is the usual radial coordinate,

$$\psi = (x^2 + y^2 + z^2)^{1/2}. \quad (5.3)$$

Solution. For the null geodesic, $ds = 0$. Therefore, we have

$$0 = -\frac{d\tau^2}{d\lambda^2} + a^2(\tau)\frac{d\psi^2}{d\lambda^2} \implies \Delta\psi = \int_{\tau_1}^{\tau_2} \frac{d\tau}{a(\tau)}. \quad (5.4)$$

- b) In the dust-filled spherical model, prove that a ray emitted at the big bang completes exactly one circuit of the universe by the big crunch.

Solution. For the dust-filled spherical model, we have

$$a = \frac{1}{2}C(1 - \cos \eta), \quad \tau = \frac{1}{2}C(\eta - \sin \eta). \quad (5.5)$$

The integral becomes

$$\Delta\psi = \int_0^{2\pi} d\eta = 2\pi. \quad (5.6)$$

- c) In the radiation-filled spherical model, prove instead that such a ray covers exactly half the universe before the big crunch.

Solution. In this case, we have

$$a = \sqrt{C'}[1 - (1 - \tau/\sqrt{C'})^2]^{1/2}, \quad (5.7)$$

which gives

$$\Delta\psi = \int_0^{2\sqrt{C'}} \frac{d\tau}{\sqrt{C'}[1 - (1 - \tau/\sqrt{C'})^2]^{1/2}} = \pi. \quad (5.8)$$